Where do libraries come from, and where do they stop helping?

Bootstrapped libraries, documentation, evidence selection, and boundaries

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In one sentence

A 4-bit 8B model wrote a verified library for a new benchmark from three examples, and a 0.6B caller that never saw the benchmark uses it correctly to 1M tokens; the useful capacity is in the proposer and in choosing evidence, not in readers, recursion or larger controllers; and on whole-document reasoning (LongBench-v2) libraries stop helping.

Libraries can be bootstrapped: an 8B wrote MRCR's library.

MRCR held out (8k to 1M tokens)	2 needles	4 needles
Direct answer, 4B	0.23	0.15
RLM, untrained 4B, no library	0.18	0.11
8B-written library, run alone	1.000	1.000
0.6B caller + library	**1.000**	**1.000**
Untrained 4B + library	0.95	0.88

64 rows per column. The 0.6B caller never saw MRCR. The proposer saw the contract and three solved examples; candidates run on 30 teaching tasks; the best scored 30/30 and was reviewed before use (40 lines of parsing, nothing copied from the examples).

Who can write it (seeds accepted)

4B Instruct: 0 of 1
4B, thinking off / on: 0 of 1 / 0 of 2
8B, thinking off: 0 of 2
8B, thinking on: **1 of 3**

Caveat: MRCR is solvable by a short parser, so this measures program discovery, not reading. We found no published note saying so.

Ledger E21, E21b; runs e21-*. MRCR: openai/mrcr (MIT), official SequenceMatcher grader.

Discovery on 37 generated families: thinking and scale gate it.

Proposer	Solved / 37	Tokens
4B Instruct (2 seeds)	5, 4	4k
8B, no thinking	4	4k
4B, thinking (30 done)	13 / 30	
8B, thinking	**20**	49k
4B + SFT on 8B's code (2 seeds)	9, 7	3k
4B + SFT round two + RL (2 seeds)	**16, 14**	2k

Tokens: proposer output per family. Families vary record format (pipe, kv, csv, prose, XML-like, chat), question and answer format; held-out families use a new format or a new question.

Training the proposer: SFT on the 8B's verified code, then 40 steps of RL with the verified teaching score as reward: 9 to 30 of 74 attempts ($p < 0.001$); string formats 1 to 22 of 42. MRCR still fails.

Helper ceiling: the 8 families that need the helper's judgment score 0.47 to 0.83, not 0.9: the 4B labels items at 0.973, and an exact total over 30 items fails often.

Ledger E26, E26b; runs e26-disc-*. Generator: family_gen.py (procedural, answers exact).

One line of documentation decides whether an untrained caller uses a library.

MRCR, untrained 4B caller, docstring	Score
Signature only	0.18
Description + signature (as the 8B wrote it)	0.59
Signature + usage line	0.95
Description + signature + usage line	0.95
Hand-written library, usage line removed	0.22
Hand-written library, as written	0.95

Usage line: `Typical use: FINAL_TEXT(answer_mrcr(request)).`

Without it the untrained 4B hands the whole input to a sub-model (overflowing at 128k and above) or retypes the routine's output by hand.

The distilled 0.6B scores 1.000 without the line; the untrained 0.6B fails either way (0.21, 0.13). Robustness comes from distillation, not size. The bootstrapper now inserts the line.

Ledger E25c, E25d; runs e25c-*, e25d-*. One family, one library, one seed per arm.

MuSiQue in long haystacks: capacity belongs in choosing evidence.

Answer F1 (90 to 180 held-out questions per row)	8k	32k	128k	tokens
Direct answer, 4B	0.52	0.25	overflow	
RLM without a library, depth 0 / depth 1	0.49 / 0.51	0.25 / 0.25	overflow	
Library that hands each hop to a child RLM	0.29	0.29	0.34	3 to 4 times
0.6B caller + filter, BM25, read	0.52	0.31	0.46	57k
0.6B caller + two-round retrieval	**0.53**	**0.39**	**0.50**	10k

What helps: a second, targeted retrieval (BM25 for the question, the helper names the missing fact, BM25 again, one read): +0.058 over BM25 alone, +0.045 over filter+BM25 at a sixth of the tokens.

What does not: recursion; an 8B reader (+0.01; gold paragraphs reach 0.63 to 0.67 with either reader); more sampling (majority of 3 at 88k tokens: 0.22 vs 0.35 for the library at 32k).

Ledger E22c to E22f, E23, E25e, E28. MuSiQue (CC BY 4.0) dev questions in haystacks of other questions' paragraphs.

Boundaries, and what we withdrew.

LongBench-v2 (150 questions, chance 0.25)

Direct, truncated to 48k tokens: 0.367
Generic map-reduce library: 0.360 (9 times the tokens)
RLM without a library: 0.120

Libraries pay where a task splits into local judgments; whole-document reasoning does not.

Withdrawn: "the reader is distractor-limited"; length routing (pre-registered: the ordering reversed on fresh questions).

Method lessons

- Verify under the semantics you run under: the kernel's `ask` raises on a cut-off reply, scripted harnesses returned partial text (one routine crashed in 46 of 90 caller runs).
- Verification must span deployment lengths: 8k-only teaching selected a pass-through routine that scores 0 at 128k.
- Routine evolution (GEPA-style): prompt rewrites by the 8B hurt; numbers helped (+0.048, ns); random search found nothing.

Ledger E32, E31, E22b, E24, E24c. LongBench-v2: zai-org (Apache-2.0).

What this changes, and what to test next.

The thesis, sharpened. For small recursive models, capability lives outside the weights: verified routines (bootstrapped by a capable proposer, or distilled into a small one), documentation that a thin caller can act on, and evidence selection inside the routine. Training pays where verification is local: in the proposer and in the caller's call discipline.

Next

1. Second distillation round: more of the 8B's libraries plus the model's own verified successes, chat included; two discovery seeds per arm.
2. Bootstrap MuSiQue on mixed-length teaching (running) and against the two-round reference.
3. Selection memory with pooled priors, since 30-question cells choose by noise.
4. A short note on program-solvable long-context benchmarks (MRCR; candidates: Michelangelo, RULER).

Appendix: protocol and provenance.

Models: Qwen3-4B-Instruct-2507 (helper, caller), Qwen3-0.6B + LoRA (distilled caller), Qwen3-4B hybrid and Qwen3-8B-AWQ (proposers), all served with vLLM 0.27.1 on two A100 MIG 7g.40gb slices.

Data: OpenAI MRCR (MIT; within-bin split, 30 teaching rows, 64 held out per needle count); MuSiQue v1.0 (CC BY 4.0; teaching from train, held out from dev; two samples of 90); LongBench-v2 (Apache-2.0, revision 2b48e494); generated families (family_gen.py, exact answers).

Statistics: paired differences with standard errors over all questions; exact McNemar for family counts; the routing test was pre-registered.

Every number above has a ledger row in docs/RESULTS.md and a run directory in the run store; the lab notebook (docs/notebook/2026-10-19-...) records failures and corrections in time order.